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compact1, compact2, compact3

java.util.concurrent.atomic

Class AtomicBoolean

java.lang.Object
    java.util.concurrent.atomic.AtomicBoolean

All Implemented Interfaces:

Serializable

```
public class AtomicBoolean
extends Object
implements Serializable
```

A `boolean` value that may be updated atomically. See the `java.util.concurrent.atomic` package specification for description of the properties of atomic variables. An `AtomicBoolean` is used in applications such as atomically updated flags, and cannot be used as a replacement for a `Boolean`.

Since:

1.5

See Also:

Serialized Form

Constructor Summary

Constructors

Constructor	Description
AtomicBoolean()	Creates a new `AtomicBoolean` with initial value `false`.

AtomicBoolean(boolean initialValue)	Creates a new AtomicBoolean with the given initial value.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
boolean	**compareAndSet**(boolean expect, boolean update)	Atomically sets the value to the given updated value if the current value == the expected value.
boolean	**get**()	Returns the current value.
boolean	**getAndSet**(boolean newValue)	Atomically sets to the given value and returns the previous value.
void	**lazySet**(boolean newValue)	Eventually sets to the given value.
void	**set**(boolean newValue)	Unconditionally sets to the given value.
String	**toString**()	Returns the String representation of the current value.
boolean	**weakCompareAndSet**(boolean expect, boolean update)	Atomically sets the value to the given updated value if the current value == the expected value.

Methods inherited from class java.lang.Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, wait, wait, wait

Constructor Detail

AtomicBoolean

```
public AtomicBoolean(boolean initialValue)
```

Creates a new AtomicBoolean with the given initial value.

Parameters:

initialValue - the initial value

AtomicBoolean

```
public AtomicBoolean()
```

Creates a new AtomicBoolean with initial value false.

Method Detail

get

```
public final boolean get()
```

Returns the current value.

Returns:

the current value

compareAndSet

```
public final boolean compareAndSet(boolean expect,
                                   boolean update)
```

Atomically sets the value to the given updated value if the current value == the expected value.

Parameters:

expect - the expected value

update - the new value

Returns:

true if successful. False return indicates that the actual value was not equal to the expected value.

weakCompareAndSet

```
public boolean weakCompareAndSet(boolean expect,
                                 boolean update)
```

Atomically sets the value to the given updated value if the current value == the expected value.

May fail spuriously and does not provide ordering guarantees, so is only rarely an appropriate alternative to `compareAndSet`.

Parameters:

expect - the expected value

update - the new value

Returns:

true if successful

set

```
public final void set(boolean newValue)
```

Unconditionally sets to the given value.

Parameters:

newValue - the new value

lazySet

```
public final void lazySet(boolean newValue)
```

Eventually sets to the given value.

Parameters:

newValue - the new value

Since:

1.6

getAndSet

```
public final boolean getAndSet(boolean newValue)
```

Atomically sets to the given value and returns the previous value.

Parameters:

`newValue - the new value`

Returns:

`the previous value`

toString

```
public String toString()
```

Returns the String representation of the current value.

Overrides:

`toString` in class `Object`

Returns:

`the String representation of the current value`

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
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